



SIMATS ENGINEERING

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Course Code: CSA0504

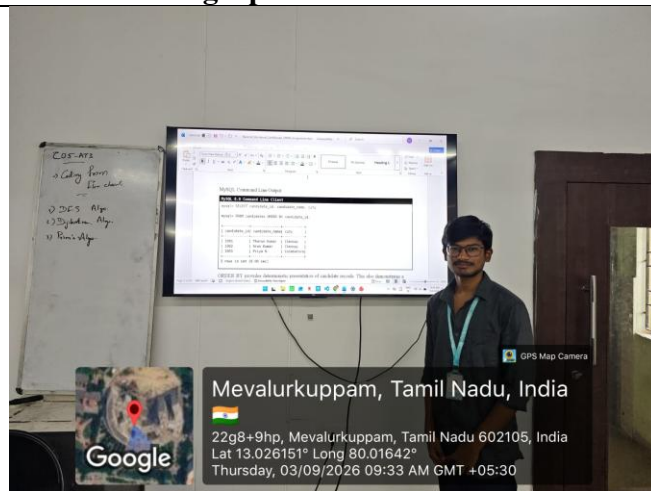
Slot: D

Course Name: Database Management Systems

Course Faculty: Dr. R. Kesavan

Project Title: Database Design and Implementation of a National Vocational Certification & Exam Slot Booking System

Module Photographs:



B. Assignment Problem / Challenge

A National Skill Certification Council conducts computer-based vocational certification exams at thousands of centers nationwide for millions of candidates each year. Every exam attempt logs question-wise responses, timing, and scoring, producing an enormous and ever-growing archive that is later used for result computation, appeals, and analytics. Candidates also book their preferred exam date, time slot, and center in real time through an online portal, and examiners publish results under strict, non-negotiable deadlines.

Two problems affect the Council's current system. First, the exam-attempt archive is stored as a single, continuously growing sequential file per year, so a query such as "retrieve candidate X's full attempt history" or "list all attempts at center Y on a given date" scans millions of records, delaying appeals and audits. Second, slot booking and result processing are implemented as un-coordinated read-then-write operations, so the same seat at a center has occasionally been booked by two candidates at once, and simultaneous result updates by different examiners have overwritten each other.

This report designs and implements a complete database solution addressing both problems using appropriate file organization, indexing and hashing for the archive, and optimized, transaction-safe operations for booking and result publishing.

Project Description:

The National Vocational Certification and Exam Slot Booking System is a database application designed to manage the complete examination process for millions of candidates across multiple exam centers. The system stores candidate details, exam attempts, booking records, center information, and result data in a structured relational database. Its primary objective is to provide fast, reliable, and secure retrieval of examination records while supporting real-time slot booking and accurate result processing. The proposed solution replaces the traditional sequential file approach with an optimized database design using indexing and hashing techniques to improve storage and retrieval efficiency.

This assignment applies advanced Database Management System (DBMS) concepts, including file organization, B+ tree indexing, hashing, SQL query optimization, transaction management, and concurrency control. The system ensures that candidate-wise and center-wise records can be retrieved within seconds, prevents double booking of exam slots, and avoids lost updates during simultaneous result processing by implementing ACID properties, COMMIT, ROLLBACK, SAVEPOINT, and appropriate isolation levels. The final solution demonstrates how efficient database engineering improves data integrity, consistency, scalability, and overall performance in a real-world national examination environment.

Student Signature

Guide Signature